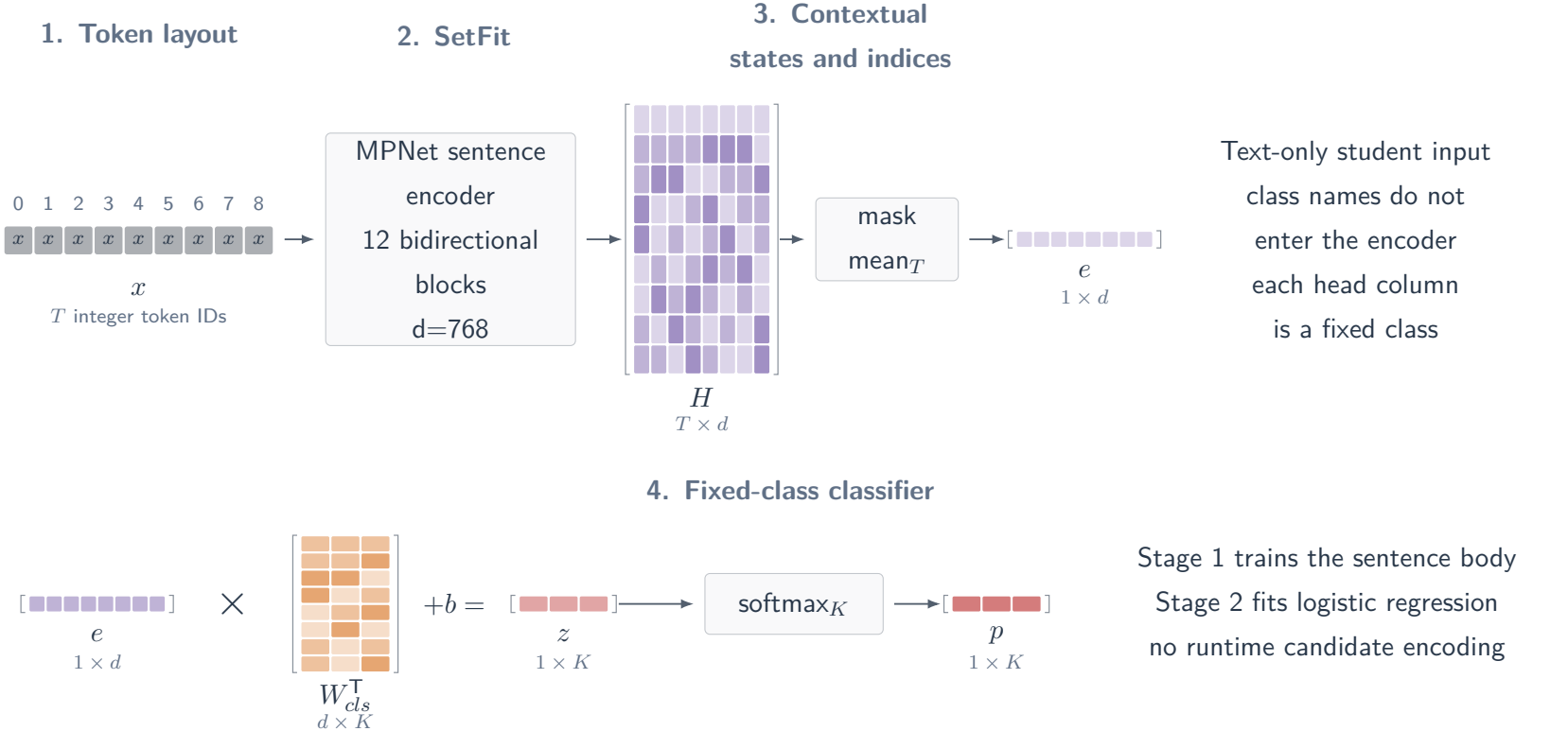
$$e = \mathrm{mean}_{mask,T}(H), \quad z = eW_{cls}^{\mathsf{T}} + b, \quad p = \mathrm{softmax}_K(z)$$



Axes	$d = 768$, K fixed; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes.
Objects	Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.
Mechanism	The selected MPNet + sklearn multiclass path has fixed labels and no forced L2 at inference. Body variants and Torch heads are separate supported modes. Phase 1: independent shared encodings and cosine MSE on same/different-class text pairs. Phase 2: fit multiclass logistic regression on sentence embeddings.